



**DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
SCHOOL OF ENGINEERING**

**CS1802 – PROGRAMMING IN PYTHON LAB
LAB MANUAL**

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Ex. 1
Date:

PYTHON IN INTERACTIVE AND SCRIPT MODE

Aim:

To explore the use of Python in the interactive and script modes in Linux.

Interactive Mode Exercises:

1. Execute the following in interactive mode:

```
width = 17  
height = 12.0  
delimiter = 'a'
```

Execute each of the following expressions and write the value of the expression and the type of the values obtained:

- *width*/2
- *width*/2.0
- *height*/3
- 1 + 2 * 5
- *delimiter* * 5

Screenshot:

2. Use the Python interpreter as a calculator to answer the following:
 - a. The volume of a sphere with radius r is $\frac{4}{3}\pi r^3$. What is the volume of a sphere with radius 5?
 - b. Suppose the cover price of a book is \$24.95, but bookstores get a 40% discount. Shipping costs \$3 for the first copy and 75 cents for each additional copy. What is the total wholesale cost for 60 copies?

Screenshot:

Script Mode Exercises:

1. Given the unit price of a product and the quantity of the product sold, find the total sale, using Python in script mode.

Algorithm:

Step 1: Get the unit price and quantity of a product from the user.

Step 2: Calculate the total sale by multiplying the unit price with the quantity of the product.

Step 3: Display the result:

Program:

Screenshot of Output:

2. Write a program (script mode) to get two integers from a user, store them in variables a and b , and evaluate the following expression:

$$c = \frac{(a + b)^2 + 10}{a \times b}$$

Algorithm:

Program:

Screenshot of Output:

Result:

Thus, the use of python in the interactive mode and the script mode has been explored.

Ex. 2**EXPLORING OPERATORS AND EXPRESSIONS****Date:****Aim:**

To explore the use of operators and expressions in Python by writing programs for the following and executing them:

- a. Get the dimensions (floating point values) of a triangle, parallelogram, cylinder, cone, sphere, and rectangular prism, calculate each of their areas and display the result as a floating-point number approximated to 2 decimal places.
- b. Calculate the simple interest and compound interest, given the principal amount (P), rate of interest (R), term of deposit (T) (in years), and number of times interest is compounded in a year (n) from the user. Display the result rounded off to 4 decimal places.
- c. Calculate the salary of an employee of a company in terms of the basic pay, dearness allowance (DA), and house rent allowance (HRA). The DA and HRA are set as a certain percentage of the basic pay. Further, the company deducts 12% of the basic pay for PF. Compute the salary that would be received by an employee given the basic pay, percentage of basic pay for DA, and percentage of basic pay for HRA and print it, rounded off to the nearest integer. (Salary = Basic Pay + DA + HRA - PF)
- d. Implement the following using functions from the math module and print the results in the scientific notation, approximated to 2 decimal places. Get the values for the variables involved from the user:
 - (i) $A \cos \theta - B \sin(\theta)$
 - (ii) $A \cos(2\pi n)$
 - (iii) e^{an}
 - (iv) Euclidean distance between two points (x1, x2) and (y1, y2) (Use the formula for Euclidean distance - $\sqrt{(x1 - x2)^2 + (y1 - y2)^2}$)
 - (v) Convert an angle theta from radians to degrees.
 - (vi) Find the base 10 and base 2 logarithm of a floating-point number x.
- e. Obtain 2 decimal numbers from the user, display them in binary, octal, and hexadecimal forms, perform the logical operations (and, or, not, left and right shift) and print the results in binary and decimal forms.
- f. Find the roots of a quadratic equation of the form $ax^2 + bx + c$. Get the values of a, b, and c from the user and display the roots.

Algorithm:

(a)

Step 1:

Step 2:

Step 3:

(b)

Step 1:

Step 2:

Step 3:

(c)

Step 1:

Step 2:

Step 3:

(d)

Step 1:

Step 2:

Step 3:

(e)

Step 1:

Step 2:

Step 3:

(e)

Step 1:

Step 2:

Step 3:

Program:

(a)

(b)

(c)

(d)

(e)

(f)

Screenshot of Output:

(a)

(b)

(c)

(d)

(e)

(f)

Result:

Thus, programs have been written and executed to explore the use of operators and expressions in Python.

Ex. 3**EXPLORING CONDITIONAL STATEMENTS****Date:****Aim:**

To explore the use of conditional statements in Python by writing programs for the following and executing them:

- g. Create a simple calculator that accepts two numbers and an arithmetic operator as inputs and performs the appropriate operation on the given numbers and displays the result.
- h. Get a year from the user and check if it is a leap year and display the result.
- i. Find the maximum of three numbers, obtained from the user, using conditional statements.
- j. Obtain the marks secured by a student in Maths, Physics, Chemistry, Computer Science, and English, out of 100, and calculate their average. Check the range within which the average mark falls and display the appropriate grade. (A+ grade - 90 to 100, A grade - 80 to 90, B+ grade - 70 to 80, B grade - 60 to 70, C grade - 50 to 60, D grade - 40 to 50, F grade - less than 40)
- k. identify if a point (x,y) lies inside, outside, or on the circumference of a circle of radius "r", centered at the origin. Obtain the values of x, y, and r from the user.
- l. Obtain the lengths of the 4 sides of a quadrilateral and the angles at each corner of the quadrilateral. Verify if the dimensions represent a valid quadrilateral and if so, check whether the dimensions represent a square, a rectangle, or neither.

Algorithm:**(a)**

Step 1:
Step 2:
Step 3:

(b)

Step 1:
Step 2:
Step 3:

(c)

Step 1:
Step 2:
Step 3:

(d)

Step 1:

Step 2:

Step 3:

(e)

Step 1:

Step 2:

Step 3:

(e)

Step 1:

Step 2:

Step 3:

Program:

(a)

(b)

(c)

(d)

(e)

(f)

Screenshot of Output:

(a)

(b)

(c)

(d)

(e)

(f)

Result:

Thus, programs have been written and executed to explore the use of conditional statements in Python.

Ex. 4**EXPLORING LOOPS****Date:****Aim:**

To explore the use of loops in Python by writing programs for the following and executing them:

- m. Print the list of prime numbers between 1 and N.
- n. Print the multiplication table up to M for a number N.
- o. Print the following pattern for 2N-1 rows.

```
      1
    2   2
  3   3   3
    2   2
      1
```

- p. Find the greatest common divisor of 2 numbers obtained from the user.
- q. Find the sum of the series $1, -(1^2), 2^3, -(3^4), 5^5, -(8^6), 13^7, \dots$ up to N terms.
- r. Find the sum of the digits of a given integer, N.
- s. Find the square root of an integer, N, using Newton's method. Obtain N and the limit, L, from the user.

Algorithm:**Program:****Screenshot of Output:****Result:**

Thus, programs have been written and executed to explore the use of loops in Python.

Ex. 5**EXPLORING STRINGS****Date:****Aim:**

To explore strings in Python by writing programs for the following and executing them:

- t. Given a string, find the list of palindromic sub-strings without using built-in functions.
- u. List the number of sub-strings with the same first and last characters in a given string.
- v. Break a given string into multiple lines with N characters each. Get the value of N from the user.
- w. Sort the characters of a string based on their frequency of occurrence.
- x. Concatenate two strings of the same length, without using built-in functions, such that the characters of each string are placed alternately in the resultant string.
- y. Encode a given string using Caesar's cipher, where each letter in a string is replaced by a letter some fixed number of positions down the alphabet. Get the alphabetic string and the number of places each letter is to be shifted from the user.

Algorithm:**Program:****Screenshot of Output:****Result:**

Thus, programs have been written and executed to explore strings in Python.

Ex. 6**EXPLORING USER-DEFINED FUNCTIONS****Date:****Aim:**

To explore user-defined functions in Python by writing programs for the following:

- z. Write a function that can generate Pythagorean triplets up to N. Obtain N from the user and pass it as an argument to your function.
- aa. Write a function to generate the sum of the sine series, $\sin(x)$ and another to generate the sum of the cosine series $\cos(x)$ up to N terms. These functions must have two arguments, x and N, where N is an optional argument. Here x is the angle in radians. Get the angle in degrees from the user and convert to radians. Pass this angle to the functions. The computation of the sum of the sine series also requires the computation of the factorial of a number, so write a function to compute factorial as well.
- bb. Write a menu-driven program to perform the following tasks. Write a separate function for each task.
 - i. Check if a given integer is an Armstrong number.
 - ii. Check if a given integer is a Deficient number.
 - iii. Check if a given integer is a Palindrome.
 - iv. Exit the program.

Algorithm:**Program:****Screenshot of Output:****Result:**

Thus, programs have been written and executed to explore user-defined functions in Python.

Ex. 7

EXPLORING RECURSION

Date:

Aim:

To explore recursions in Python by writing recursive functions for the following:

- cc. Find the factorial of a given integer.
- dd. Find the value of a number, x, raised to the power of another number, y.
- ee. Write a menu driven program to generate the following sequences up to N terms by writing a recursive function for each:
 - (i) Fibonacci series
 - (ii) Arithmetic progression, given the starting number and the common difference
 - (iii) Geometric progression, given the starting number and the common ratio
 - (iv) Harmonic progression, given the starting number and the common difference
 - (v) Triangular number series
- ff. Display the binary equivalent of a decimal number.
- gg. Find the largest digit in a given integer.
- hh. Compute nC^r .
- ii. Display the sequence of steps to be followed to solve the Tower of Hanoi problem, given the number of disks, N.

Algorithm:

Program:

Screenshot of Output:

Result:

Thus, programs have been written and executed to explore recursive functions in Python.

Ex. 8**EXPLORING LISTS AND TUPLES****Date:****Aim:**

To explore lists and tuples in Python by writing programs for the following:

- jj. Sort a list of numbers obtained from the user using Bubble sort, without using built-in functions. Find the kth largest number in the list. Obtain k from the user.
- kk. Sort a list of numbers obtained from the user using Insertion sort, without using built-in functions.
- ll. Perform a linear search on a given list to find a given element and return its position if present in the list, without using built-in functions.
- mm. Use the binary search algorithm to check if a given element is present in a given tuple, without using built-in functions, except the “sorted” function.
- nn. Count the frequency of occurrence of each element of a tuple and display the elements and their count. Find the elements that appear only once in the tuple. (Write 2 programs, one that makes use of tuple methods and functions, and another that does not.)

Algorithm:**Program:****Screenshot of Output:****Result:**

Thus, programs have been written and executed to explore lists and tuples in Python.

Ex. 9**EXPLORING NESTED LISTS****Date:****Aim:**

To explore nested lists in Python by writing programs for the following:

- oo. Obtain 2 matrices from a user in the form of nested lists and perform matrix multiplication and addition, without using built in functions.
- pp. Given the names and grades for each student in a class, store them in a nested list and print the name(s) of any student(s) having the second lowest grade.
- qq. Check if a given list contains nested lists.
- rr. Find the list in a list of lists whose sum of elements is the highest.

Algorithm:**Program:****Screenshot of Output:****Result:**

Thus, programs have been written and executed to explore nested lists in Python.

Ex. 10**EXPLORING DICTIONARIES****Date:****Aim:**

To explore dictionaries in Python by writing programs for the following:

- a. Obtain a string from the user and create a dictionary that contains each character in the string mapped to the number of times the character appears in the given string. Write programs with and without the count method.
- b. Morse Code defines a standard encoding where each letter is mapped to a series of dots and dashes. Assume that the list of codes for each English alphabet is as follows:
[".-","-...","-.-","-..",".", "-.-","--.", "...", ". .", "---", "-.-", ".-.", "--", "-.", "---", "-.-", "--.-", ".-.", "...", "-.", "-.-", "...-", ".--", "-.-", "-.-", "-.-"]
Create a dictionary that maps each alphabet to the corresponding Morse code. Obtain a list of words from the user and display the Morse code corresponding to each word. Store the codes in a dictionary. If the user inputs the same word multiple times, fetch the corresponding code from the word dictionary instead of computing the code again. Find out if multiple words have the same Morse code and display them.
- c. Obtain the marks secured by N students in 5 subjects namely, Math, English, Physics, Chemistry, and Computer Science from the user. The marks secured by each student may be received in the form of a list from the user and stored in a dictionary. Using the data obtained, create a nested dictionary in the following format, which contains the marks, average mark, and rank of each student:

{Student1: {Marks: [50, 45, 36, 48, 50], Average: 45.8, Rank: 2}, Student 2: {Marks: [43, 44, 48, 30, 37], Average: 40.4, Rank: 3}, Student3: {Marks: [50, 46, 50, 50, 50], Average: 49.2, Rank: 1}}
- d. Create a dictionary grouping a sequence of key-value pairs into a dictionary of lists.
- e. Read N numbers from a user and create a dictionary having two keys, EVEN and ODD which point to a list of even and odd numbers respectively.

Algorithm:**Program:****Screenshot of Output:****Result:**

Thus, programs have been written and executed to explore dictionaries in Python.

Ex. 11

EXPLORING FILE OPERATIONS

Date:

Aim:

To explore file operations in Python by writing programs for the following:

- f. Create a new text file and write 10 lines into it.
- g. Read the text file created above and count the number of words, characters, and lines in it. Find the frequency of occurrence of each word and character.
- h. Read only the first 2 lines of the text file created above and print them. Then skip the next 4 characters and print the remaining characters in the line. Then, skip 3 lines and print the remaining lines.
- i. Create a text file with several lines. Compute the unigram, bigram, and trigram probabilities (that is, probability of occurrence of each word, pair of words, and combination of 3 words). Remove all punctuation marks from the text before computing the probabilities. Store the computed probabilities in dictionaries and display them.

Algorithm:

Program:

Screenshot of Output:

Result:

Thus, programs have been written and executed to explore file operations in Python.

Ex. 12**EXPLORING COMMAND-LINE ARGUMENTS****Date:****Aim:**

To explore command-line arguments in Python by writing programs for the following:

- j. Given a sentence as a command-line argument, compute the probability of the sentence being valid based on bigram probabilities computed in the previous exercise.
- k. Given a file name as a command line argument, check if the file exists. If the file exists, read the contents of the file and display it. If the file does not exist, create the file and add some content to it. (Use try ... except ... else blocks.)
- l. Given an integer as a command-line argument, print the list of prime numbers less than the given integer.

Algorithm:**Program:****Screenshot of Output:****Result:**

Thus, programs have been written and executed to explore command-line arguments in Python.